(KTXFI1EBNF) Physics I. Lecture

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March 2, 2026

Current Updates and Issues

- The materials for the previous lecture and practice session have been uploaded to Moodle, YouTube, and GitHub.
- I have also added short videos to Moodle to help clarify the underlying physics.
- We will conclude the Mechanics module today; consequently, Homework No. 1 will be released shortly (Deadline: March 23, 2026).
- I am still awaiting my contract from the International Space University. My tentative start date is March 23, 2026.

Requirements I

Students who fully comply with all of the points listed below during the semester may get a signature:

- ~~The number of absences from lectures and exercises during the semester did not exceed the limit specified in the TVSZ (maximum 30% of the total number of hours held).~~
- Correctly completed 4 3 of the ~~6~~ 5 assigned homework assignments, submitted on time and in the required format.
- Achieved a score of at least 40% on the midterm exam.

Requirements II

- Extra credit: In addition to the homework assignments submitted during the semester, two more assignments can be submitted as extra credit. If the result of the assignment submitted in this way is correct, the student will receive $+10$ semester points per assignment. In this way, a maximum of $+20$ semester points can be earned with extra assignments, which will be included in the final semester score. ~~The extra credit assignment can be submitted either during the two weeks of the given homework assignment or, at the latest, during the week designated for making up homework assignments (see below for make-up assignments in the 13th week of instruction).~~

Requirements III

- Midterm exam: A poorly written or incorrect midterm exam with a score above 40% cannot be corrected, i.e., it cannot be retaken for the purpose of correction. There is no retake exam. The score received for the extra credit assignment cannot be added to the score obtained on the midterm exam, i.e., the 40% midterm exam score must be achieved solely from the midterm exam assignments.
- Grades: Insufficient/Fail (1): 0-50 %, Sufficient/Pass (2): 51-60 %, Average (3): 61-75 %, Good (4): 76-90 %, Excellent (5): 91- %.

Recommended

Richard P. Feynman: The Feynman Lectures on Physics, Vol. I: The New Millennium Edition: Mainly Mechanics, Radiation, and Heat, New York, 2010, ISBN-10 0465024939, IBN-13 978-0465024933

Where are we? I

1. **Mechanics I.:** Newton's laws, equation of motion (review). Momentum. General concept of mechanical work, work theorem. Forms of mechanical energy.
2. **Mechanics II.:** Weight force, gravitational force, gravitational field. Work of the gravitational force. Conservative force field. Dissipative force field. Mechanical power.
3. **Mechanics of Rigid Bodies I.** Concept of a rigid body. Forces acting on a rigid body. Motion of a rigid body. Concept of the centre of mass. Location of the centre of mass. Torque. Angular momentum. Energy of a rotating rigid body, angular momentum of a rotating rigid body, Steiner's theorem.
4. **Mechanics of Rigid Bodies II.** Principal theorems of rigid body motion; conservation laws. Collisions.
5. **Oscillations I.** Dynamic condition of harmonic oscillatory motion. Differential equation of harmonic undamped oscillatory motion.

Where are we? II

6. **Oscillations II.** Superposition of oscillations. Decomposition of oscillations into components. Damped oscillations. Forced oscillations, resonance. Coupled oscillations.
7. **Wave phenomena:** *Reflection of waves, refraction of waves, diffraction of waves. Huygens–Fresnel principle. Interference of waves. Polarization of waves.*
8. **Elements of acoustics:** *Basics of acoustics. Fundamental concepts. Doppler effect. Sound intensity, loudness according to Weber–Fechner's law.*
9. **Thermodynamics I:** Temperature scales, thermal expansion, thermometers. State variables, ideal gas, state changes, ideal gas equation of state, Boyle's law, Gay-Lussac's laws, universal (combined) gas law, heat, specific heat, molar heat, heat capacity.
10. **Thermodynamics II:** Expansion work, internal energy, the first law of thermodynamics, forms of the first law in different state changes.

Where are we? III

11. **Thermodynamics III:** Cyclic processes, Carnot cycle, Carnot's theorem, reduced heat, entropy, the second law of thermodynamics.
12. **Thermodynamics IV:** Kinetic theory of gases, molecular thermodynamics. Distributions, distribution functions. Elementary statistics.
13. **Electrostatics I.:** Electric charge: Fundamental electrical phenomena, electric charge and matter, Coulomb's law (force between two point charges, unit of electric charge, permittivity of vacuum).
14. **Electrostatics II.:** The electric field: The electrostatic field (electric field strength, electric field lines). Determination of electric field strength. Gauss's law (flux of the electric field). Electric field of a point charge.

Where are we? IV

15. **Electrostatics III.:** Electrostatic potential: Work of the electric field (work done in an electric field). Electric potential (electrostatic potential). Electric potential difference, voltage. Electric potential of charge systems at small and large distances. Electrostatic energy.
16. **Elements of physical or wave optics.** Branches of optics. Light interference, light diffraction, diffraction through a slit, optical grating, light polarization. Light as an electromagnetic wave: Poynting vector. Optical instruments.

Fundamental Wave Quantities I

- Mathematical Description of a Harmonic Wave: a harmonic traveling wave propagating in the $+x$ direction can be written as:

$$y(x,t) = A\sin(kx - \omega t + \varphi)$$

  where A is amplitude, k is wave number, ω is angular frequency, and φ is initial phase.
- This expression describes both spatial and temporal oscillation.
- **Definition:** (Amplitude) The amplitude A is the maximum displacement of the oscillating quantity from its equilibrium position.
- From the wave equation:

$$y_{\text{max}} = A$$

Fundamental Wave Quantities II

- Physical meaning:
  - Determines the energy transported by the wave
  - For many waves: Energy $\propto A^2$
- Example (mechanical wave): maximum vertical displacement of a string.
- Example (electromagnetic wave): maximum electric field strength.
- **Definition:** (Wavelength) The wavelength λ is the spatial period of the wave, i.e., the distance between two successive points oscillating in phase.
- Examples of points in phase: (i) crest to crest, (ii) trough to trough.
- Relation to wave number:

$$k = \frac{2\pi}{\lambda}$$

- Therefore,

$$\lambda = \frac{2\pi}{k}$$

Fundamental Wave Quantities III

- **Definition:** (Frequency) The frequency f is the number of oscillations per unit time.
$$f = \frac{1}{T},$$
where T is the period of oscillation.
- Relation to angular frequency:
$$\omega = 2\pi f$$
- Hence,
$$f = \frac{\omega}{2\pi}$$
Unit of frequency: Hertz (Hz).
- Wave Propagation Speed: A traveling wave moves with a phase velocity v.

Fundamental Wave Quantities IV

- Condition of constant phase:

$$kx - \omega t = \text{constant}$$

- Differentiating with respect to time:

$$k\frac{dx}{dt} - \omega = 0$$

- Therefore,

$$v = \frac{dx}{dt} = \frac{\omega}{k}$$

  This is the phase velocity of the wave.

Fundamental Wave Quantities V

- Fundamental Relation Between Speed, Wavelength and Frequency
- Using

$$k = \frac{2\pi}{\lambda} \quad \text{and} \quad \omega = 2\pi f$$

- We obtain

$$v = \frac{\omega}{k}$$

- Substituting:

$$v = \frac{2\pi f}{2\pi/\lambda}$$

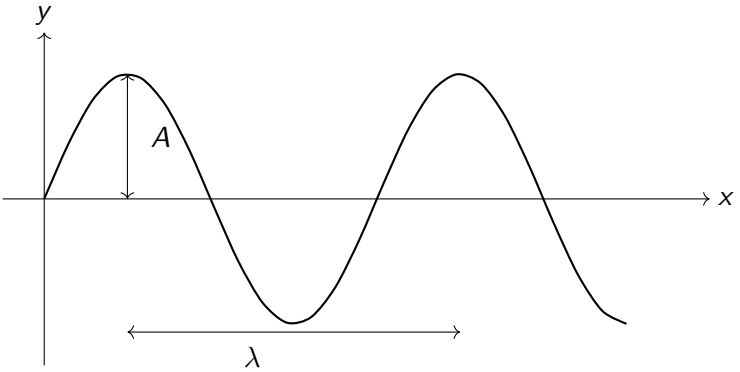
- Final relation:

$$\boxed{v = \lambda f}$$

This is the fundamental wave equation.

Fundamental Wave Quantities VI

- Graphical Representation of a Wave

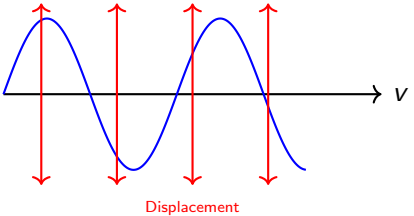


- Amplitude A: maximum vertical displacement.
- Wavelength λ: spatial repetition length.

Fundamental Wave Quantities VII

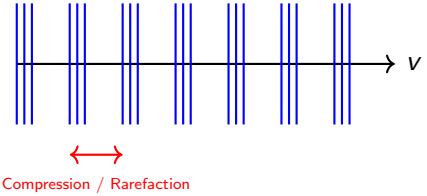
Transverse Waves

- Particle displacement is *perpendicular* to the direction of propagation.
- Examples: Light, surface water waves, S-waves in earthquakes.



Longitudinal Waves

- Particle displacement is *parallel* to the direction of propagation.
- Examples: Sound waves, ultrasound, P-waves in earthquakes.



Fundamental Wave Quantities VIII

- Physical Interpretation
  - Amplitude controls the wave energy.
  - Frequency determines how fast the system oscillates.
  - Wavelength determines spatial periodicity.
  - Propagation speed links space and time behavior.
- Changing medium:
  - Frequency remains constant.
  - Speed changes.
  - Therefore wavelength changes according to $\lambda = \frac{v}{f}$.

Fundamental Wave Quantities IX

- Summary of Key Relations

$$
\begin{aligned}
y(x,t) &= A\sin(kx - \omega t + \varphi) \\
k &= \frac{2\pi}{\lambda} \\
\omega &= 2\pi f \\
v &= \frac{\omega}{k} \\
v &= \lambda f
\end{aligned}
$$

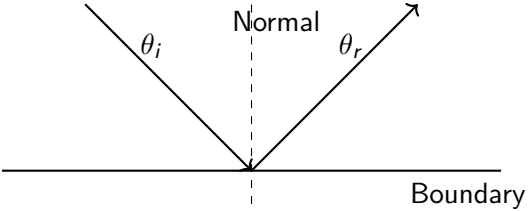
- These quantities fully characterize a monochromatic harmonic wave.

Wave Phenomena I

- **Definition:** (Reflection of Waves) Reflection is the change in direction of a wave at a boundary between two media such that the wave returns into the original medium.
- **Law of reflection:**

$$\theta_i = \theta_r,$$

  where θ_i is the angle of incidence, θ_r is the angle of reflection.
- The incident and reflected rays are symmetric with respect to the normal.



Wave Phenomena II

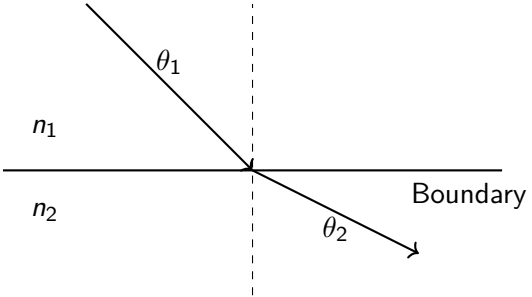
- **Definition:** (Refraction of Waves) Refraction is the change in direction of a wave when it passes from one medium to another due to a change in wave speed.
- Wave speed relation:

$$v = \frac{c}{n}$$

- Snell's law:

$$n_1 \sin\theta_1 = n_2 \sin\theta_2,$$

  where n is the refractive index.
- If $n_2 > n_1$, the wave bends toward the normal.

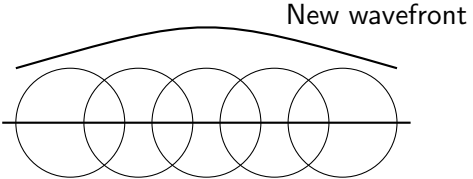


Wave Phenomena III

- **Definition:** (Huygens–Fresnel Principle) Every point of a wavefront acts as a source of secondary spherical wavelets. The new wavefront is the envelope of these wavelets.
- Mathematical idea:

$$U(P) = \int_{\text{wavefront}} K(\theta) \frac{e^{ikr}}{r} \, dS,$$

  where $k = \frac{2\pi}{\lambda}$, r is distance from secondary source, and $K(\theta)$ is obliquity factor.
- The envelope of secondary wavelets forms the new advancing wavefront.



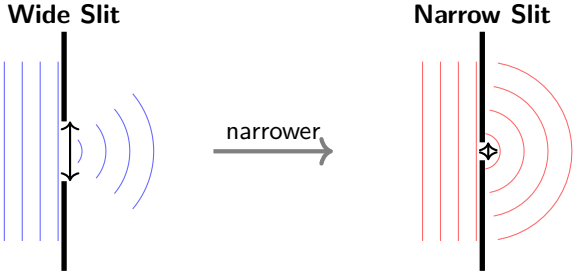
Wave Phenomena IV

- **Definition:** (Diffraction of Waves) Diffraction is the spreading of waves when they pass through an aperture or around an obstacle.
- Single-slit condition for minima:

$$a \sin \theta = m\lambda,$$

where a is slit width, λ is wavelength, and $m = \pm 1, \pm 2, \ldots$.

- Single-Slit Diffraction Pattern: Narrower slit $\Rightarrow$ stronger spreading.



Wave Phenomena V

- **Definition:** (Interference of Waves) Interference is the superposition of two or more coherent waves producing a resultant intensity pattern.
- Resultant displacement:

$$y = y_1 + y_2$$

- Constructive interference:

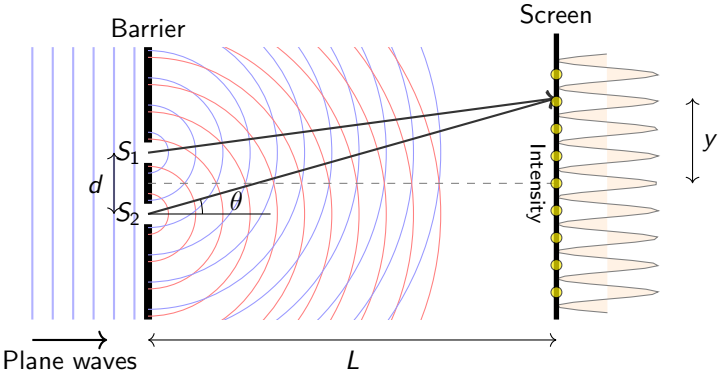
$$\Delta = m\lambda$$

- Destructive interference:

$$\Delta = \left(m + \frac{1}{2}\right)\lambda$$

Wave Phenomena VI

- Double-Slit Interference



where d is distance between the two slits (S_1, S_2), L is distance from the slits to the screen, θ is diffraction angle, and y is position of the fringe on the screen.

Wave Phenomena VII

- **Condition for constructive interference (bright fringes):**

$$d \sin\theta = m\lambda,$$

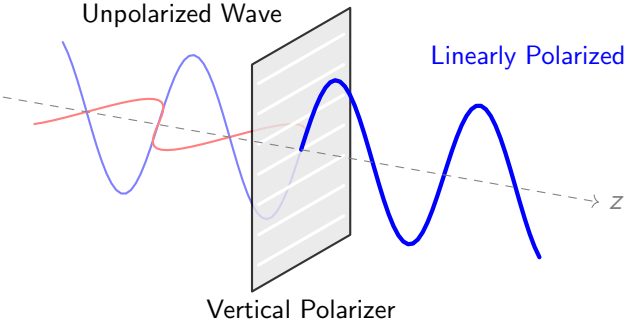
  where $m = 0, \pm 1, \pm 2, \ldots$
- For small angles, using $\sin\theta \approx \tan\theta \approx \frac{y}{L}$:

$$y_m = \frac{m\lambda L}{d}$$

- This equation gives the position of the m-th bright fringe on the screen.

Wave Phenomena VIII

- **Definition:** (Polarization of Waves) Polarization describes the orientation of oscillations in a transverse wave.



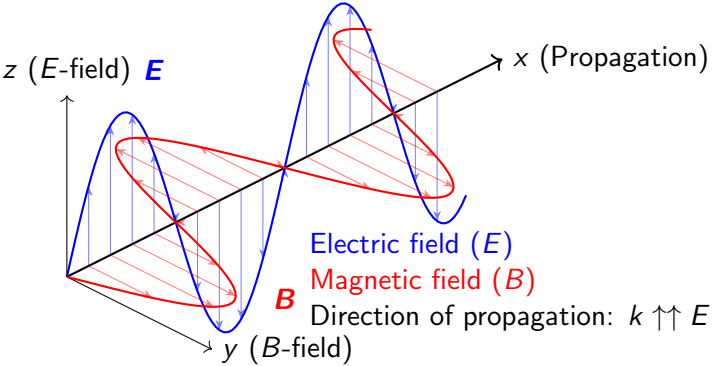
- Only transverse waves can be polarized.

Wave Phenomena IX

- For electromagnetic waves:

$$\vec{E} \perp \vec{B} \perp \vec{k},$$

where $\vec{E}$ is electric field, $\vec{B}$ is magnetic field, and $\vec{k}$ is propagation direction.

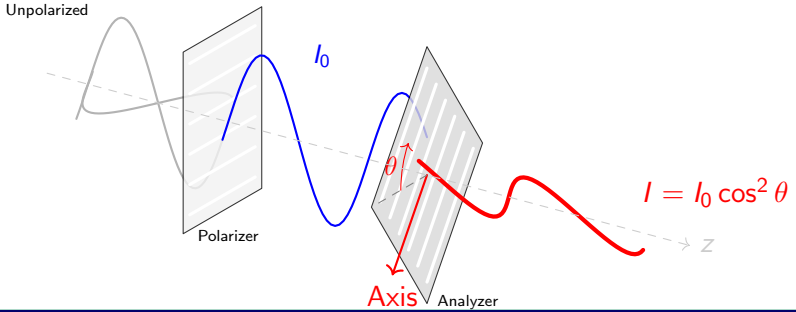


Wave Phenomena X

- **Malus' Law**: The transmitted intensity depends on the angle between the polarization direction and the analyzer axis:

$$I = I_0 \cos^2 \theta,$$

where I is the transmitted intensity, I_0 is the initial intensity of the polarized light, and θ is the angle between the polarizer and the analyzer.



Wave Phenomena XI

- Summary
  - Reflection: $\theta_i = \theta_r$
  - Refraction: $n_1 \sin\theta_1 = n_2 \sin\theta_2$
  - Diffraction: spreading due to finite aperture
  - Interference: superposition principle
  - Polarization: orientation of transverse oscillations
- All phenomena follow from wave superposition and boundary conditions.

Basics of Acoustics I

- **Definition:** Sound is a mechanical longitudinal wave that propagates through a material medium due to pressure variations.
- Characteristics:
  - Requires a medium (air, liquid, solid)
  - Cannot propagate in vacuum
  - Caused by oscillations of particles
- Sound waves in air are pressure waves.
- Longitudinal Sound Wave

- Particles oscillate parallel to the direction of propagation.

Basics of Acoustics II

- Wave Equation for Sound for a plane harmonic sound wave:

$$p(x,t) = p_0 \sin(kx - \omega t),$$

  where p_0 is pressure amplitude, $k = \frac{2\pi}{\lambda}$ is wave number, and $\omega = 2\pi f$ is angular frequency.
- Wave speed:

$$v = \lambda f = \frac{\omega}{k}$$

- Speed of Sound in gases:

$$v = \sqrt{\frac{\gamma p}{\rho}},$$

  where γ is adiabatic index, p is equilibrium pressure, and $\rho =$ density.
- In air at room temperature:

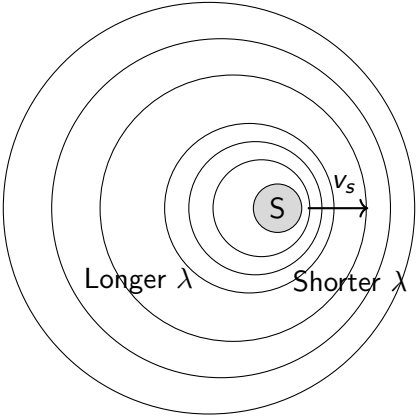
$$v \approx 343 \text{ m/s}$$

Basics of Acoustics III

- **Definition:** (Doppler Effect) The Doppler effect is the apparent change in frequency due to relative motion between source and observer.
- General formula:

$$f' = f\frac{v \pm v_o}{v \mp v_s},$$

  where v is speed of sound, v_o is observer velocity, and v_s = source velocity.
- Upper sign: motion toward each other.
- Wavelength decreases in front and increases behind the moving source.



Basics of Acoustics IV

- **Definition:** (Sound Intensity) Sound intensity is the acoustic power transmitted per unit area.

  Unit: W/m^2

$$I = \frac{P}{A}$$

- For a spherical wave:

$$I = \frac{P}{4\pi r^2}$$

- Intensity decreases with the square of distance.
- Intensity and Pressure Relation for a plane sound wave:

$$I = \frac{p_0^2}{2\rho v}$$

- Therefore,

$$I \propto p_0^2$$

- Sound intensity is proportional to the square of the pressure amplitude.

Basics of Acoustics V

- Sound Intensity Level (Decibel Scale): Human hearing responds logarithmically.
- Intensity level:

$$\beta = 10 \log_{10} \left(\frac{I}{I_0} \right),$$

where:

$$I_0 = 10^{-12} \text{ W/m}^2$$

Unit: decibel (dB)

- **Weber–Fechner law:** Perceived sensation is proportional to the logarithm of the stimulus.

$$L \propto \log I$$

- For sound:

$$L = k \log \left(\frac{I}{I_0} \right)$$

- This explains why loudness perception is logarithmic.
- Doubling intensity does not double perceived loudness.

Basics of Acoustics VI

- Summary
  - Sound is a longitudinal mechanical wave.
  - Wave speed depends on medium properties.
  - Doppler effect shifts observed frequency.
  - Intensity measures acoustic power per area.
  - Loudness follows a logarithmic perception law.
- Key relations:

$$
\begin{aligned}
v &= \lambda f \\
\beta &= 10 \log_{10} \left(\frac{I}{I_0} \right)
\end{aligned}
$$

The End

Thank you for your attention!